

NXP Semiconductors	SPT Kernels for RRU2	04/13/2021
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SPT Kernels for RRU2

Design Document

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1 Document purpose

The SPT Kernels represent sequences of instructions that implement signal processing algorithms and can be run on the SPT hardware. SPT Kernels can be classified based on the algorithms they provide (Range-FFT, Doppler-FFT etc.) and all kernels that implement the same functionality follow a common pattern.

This pattern has two interdependent components: the processing flow and the memory layout. Both of these are important to understand in order to get used to SPT programming. The focus of this document is to summarize these aspects through diagrams dedicated to each type of kernel.

2 Guidelines for interpretation

In this document, each type of kernel is described through a diagram. In order to interpret them correctly it is best to follow these guidelines:

2.1 Diagram structure

Processing flow:

- The processing flow is described as an activity diagram located in the center of each figure together with the corresponding notes positioned in the right side of each diagram. It is focused on the main instructions in each implemented algorithm, meaning that instructions with a small impact on the processed data, such as set/get on work registers, are not illustrated.
- Most kernels make use of the ASYNC feature of the PDMA. This is done by transferring one chunk of data to OPRAM with a SYNC PDMA, triggering the ASYNC PDMA for the next chunk of data and starting to process the first chunk in parallel with the PDMA. Because the 2 chunks of data are processed in the same way, there is a set of duplicated operations that correspond to the 2 PDMA transfers. This **correspondence** is illustrated through some light-yellow arrows.
- For use cases with MIMO, compression or adaptive scaling, the processing flow branches accordingly to cover all the possibilities. Each type of use case is described by a keyword:
 - for MIMO specific actions, the keyword is **MIMO**
 - for compression use cases, the keyword is **CP4Q**
 - for adaptive scaling, in the range kernel the associated keyword is **ADPTV**, while in the doppler kernel **RCOMP** is used

Memory layout:

- The memory layout located on the left side of the diagram focuses on how data flows through memory. Each **significant** change in the memory layout is placed in alignment with the instruction that triggered it.
- Memory diagrams are **not** drawn for instructions that only change the **values** placed in memory and not its layout.
- Because there are kernels for both 4 and 8 channels, some memory blocks are **duplicated** in order to display the memory layout for both cases. These blocks have a comment above with the name of the instruction that produced the layout.
- For the kernels that use the ASYNC PDMA as described above, the duplicated instructions are performed on **different** OPRAM banks. The memory blocks usually describe only one subset of OPRAM banks, used by the first set of instructions. The layout is the same for the second set of instructions.
- Some memory diagrams are not the direct result of an instruction. These are usually presented in order to illustrate how the memory looks like at the end of a loop. They are connected to their corresponding point in the activity diagram through a yellow dotted line.

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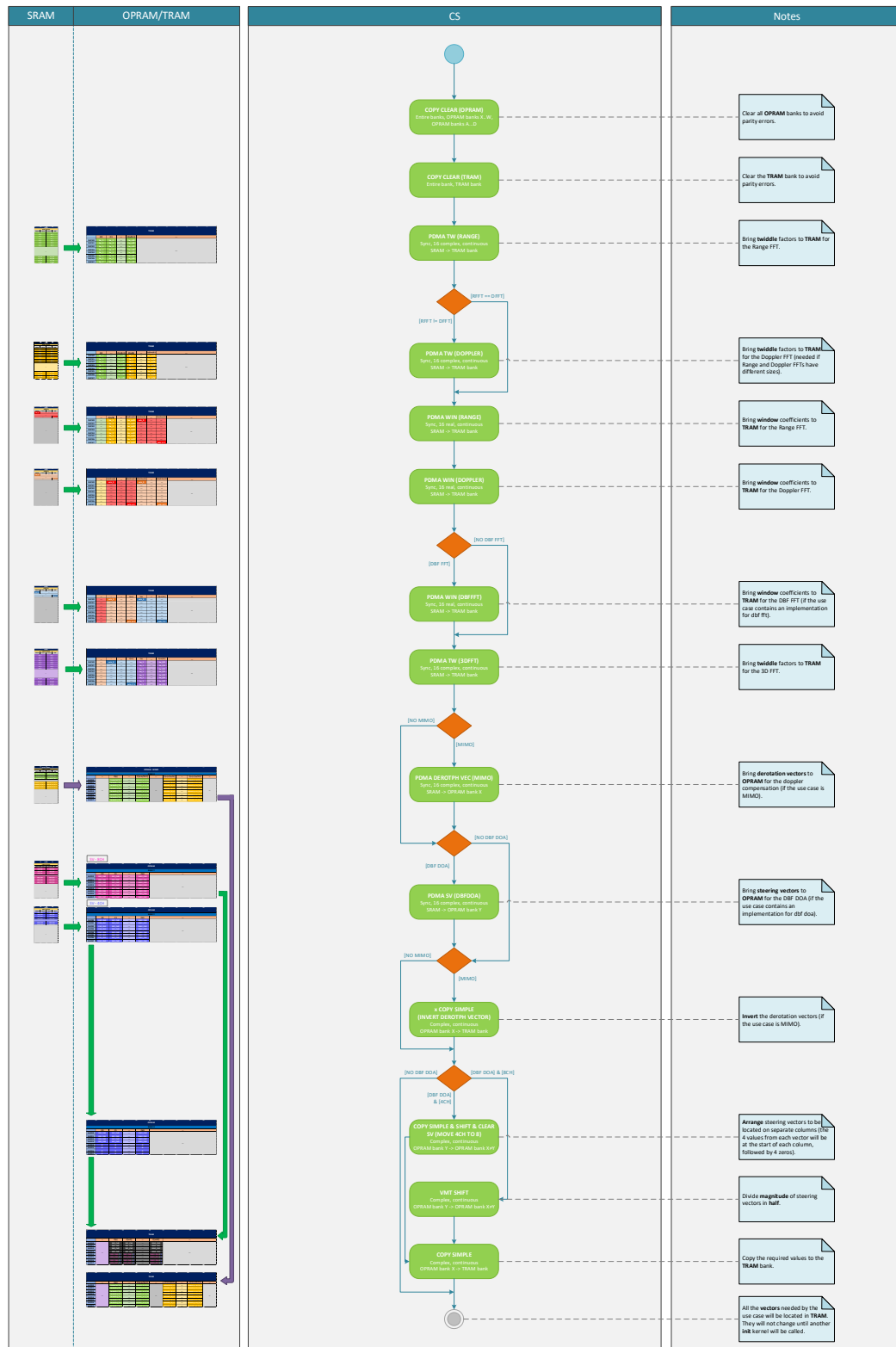
- In all diagrams, the transitions in the memory layout follow a color code:
 - green arrows describe the generic transitions, applicable to all kernels
 - purple arrows correspond to MIMO use cases
 - orange arrows display CP4D specific actions
 - blue arrows describe transitions for adaptive scaling use cases.

2.2 Notations

- In all diagrams, **n** represents the number of samples and **m** represents the number of chirps. The number of range bins is **n/2** and the number of doppler bins is also **m**.
- In diagrams that include MIMO use cases, **s** represents the number of TX slots.
- In diagrams that describe FFTs with zero padding, **t** represents the size of the oversampled FFT.
- In the description of the processing flow, for kernels that use ASYNC PDMA, OPRAM banks used by the first set of instructions are noted as **X/Y/Z/W** and the ones used by the second set of instructions are noted as **A/B/C/D**. This is done in order to emphasize that they are, most of the time, different banks that are used for different sets of data.

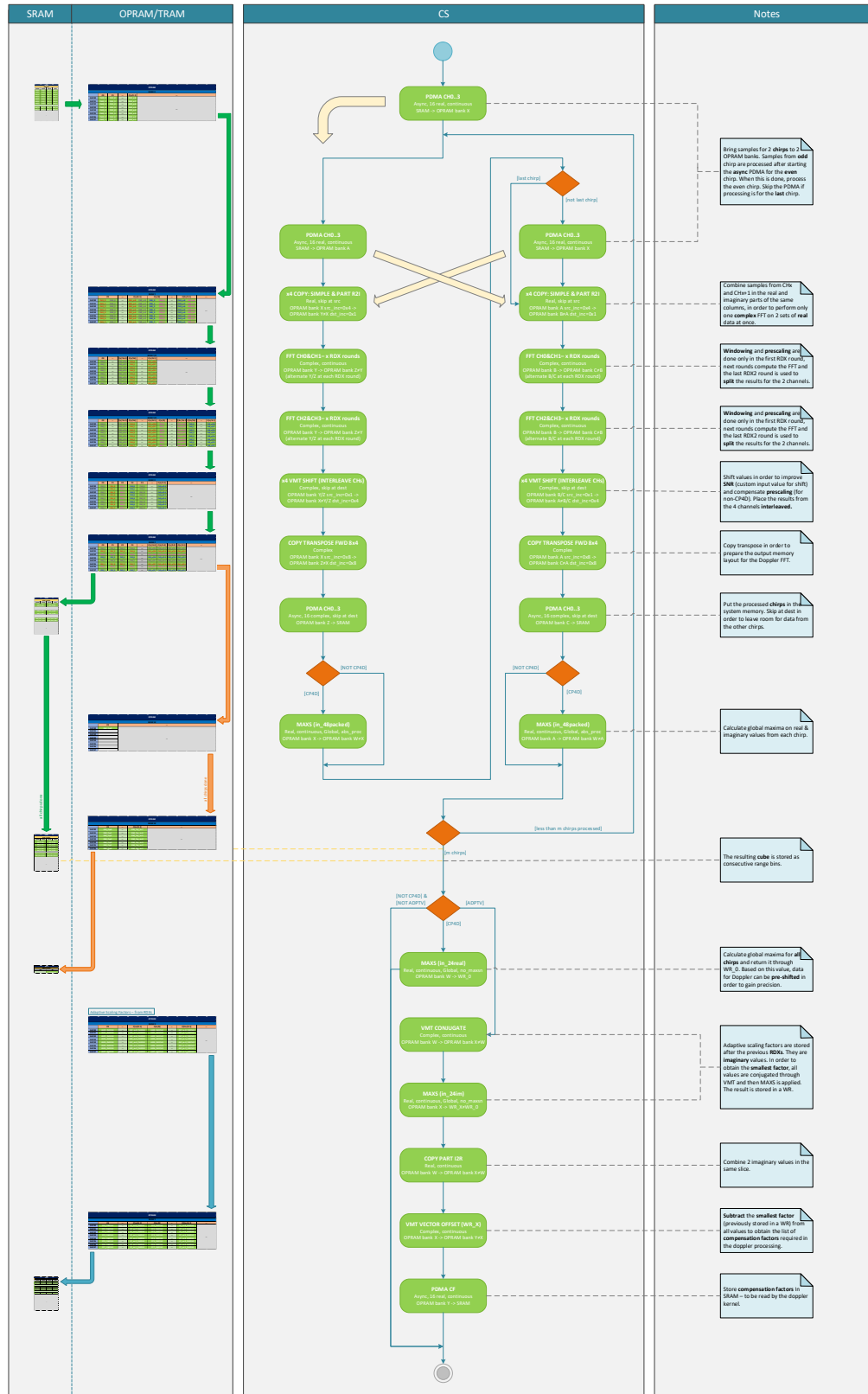
3 SPT Kernels Design

3.1 Init Kernel

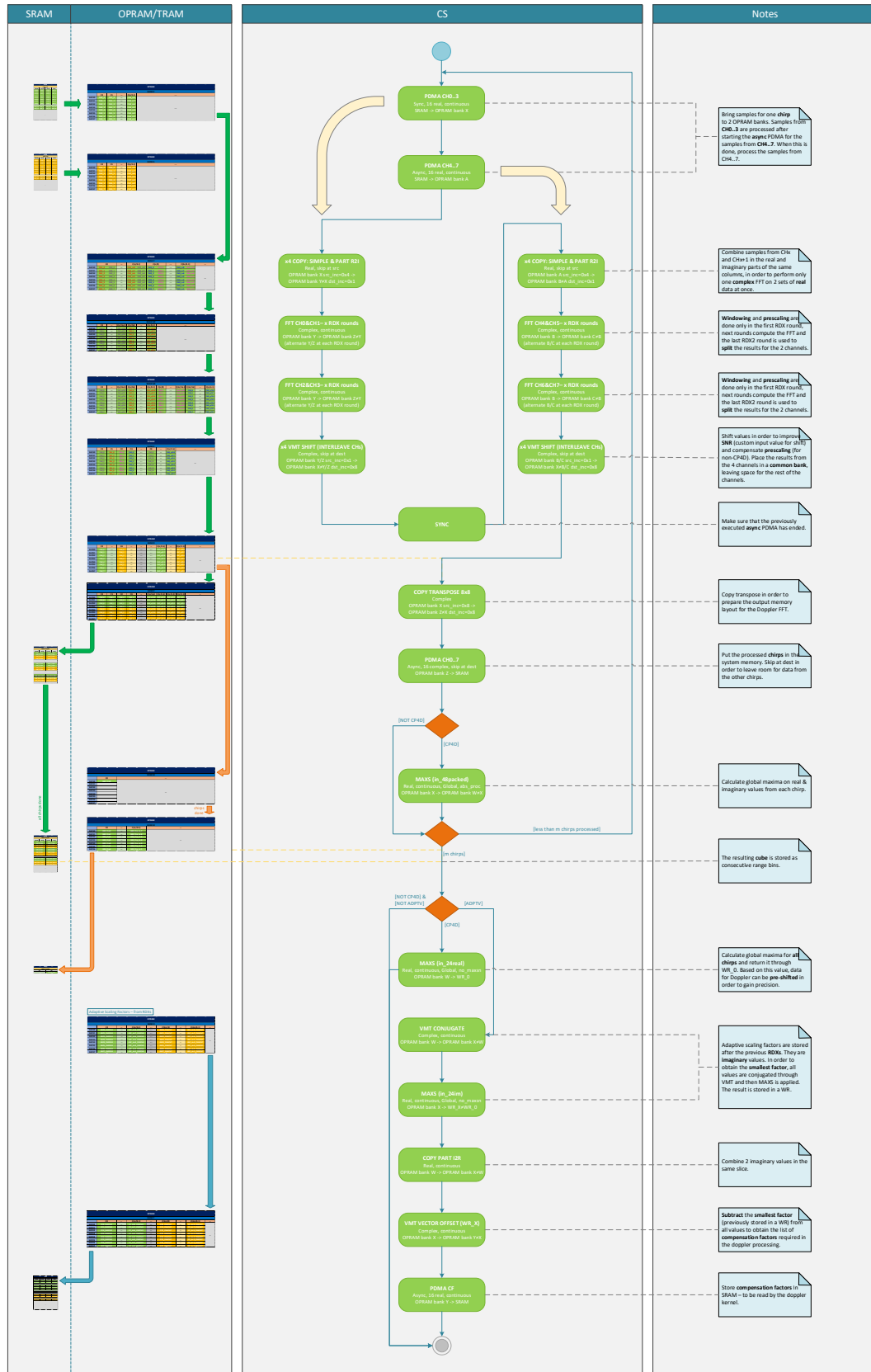


3.2 Range Kernel

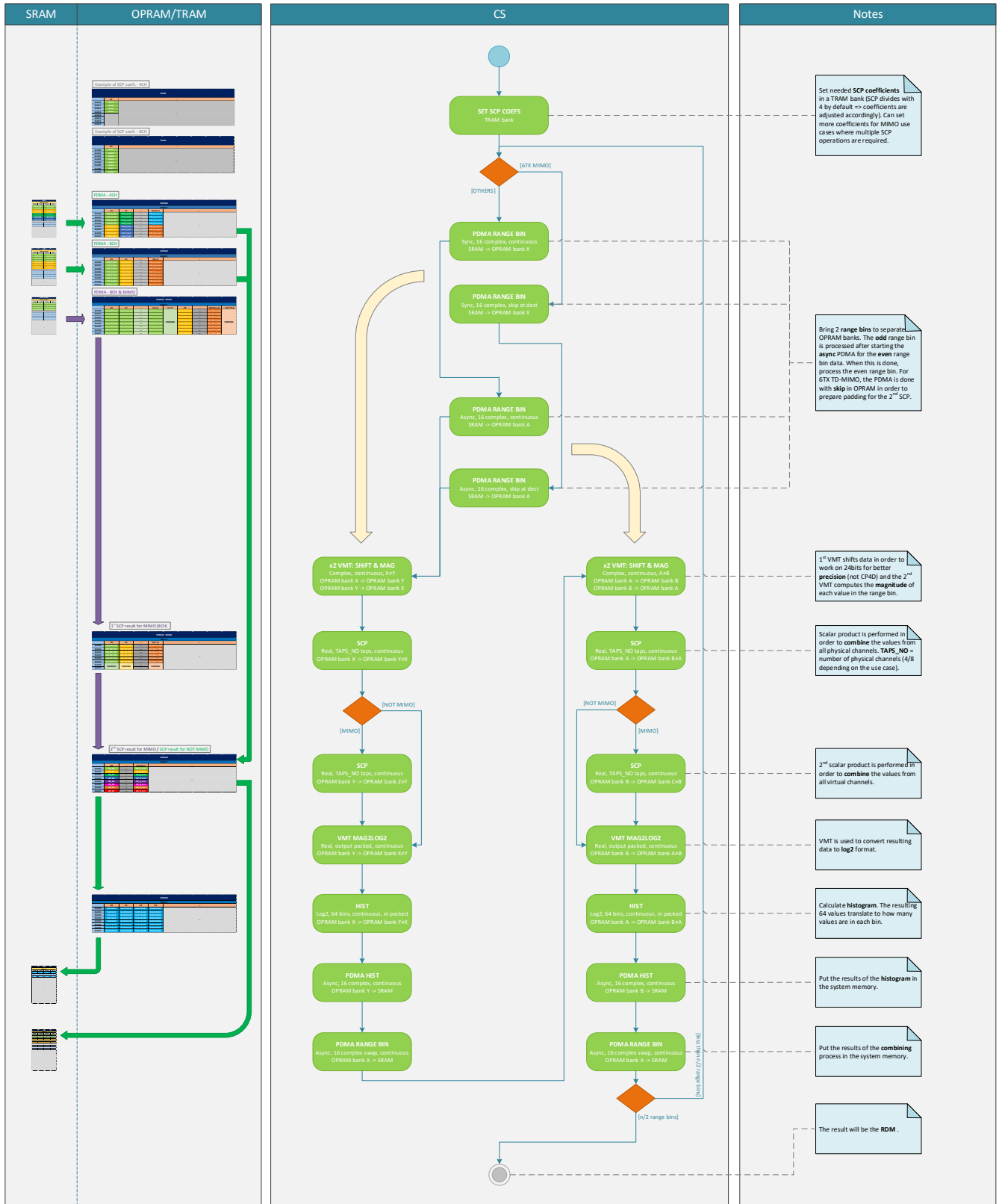
3.2.1 4CH



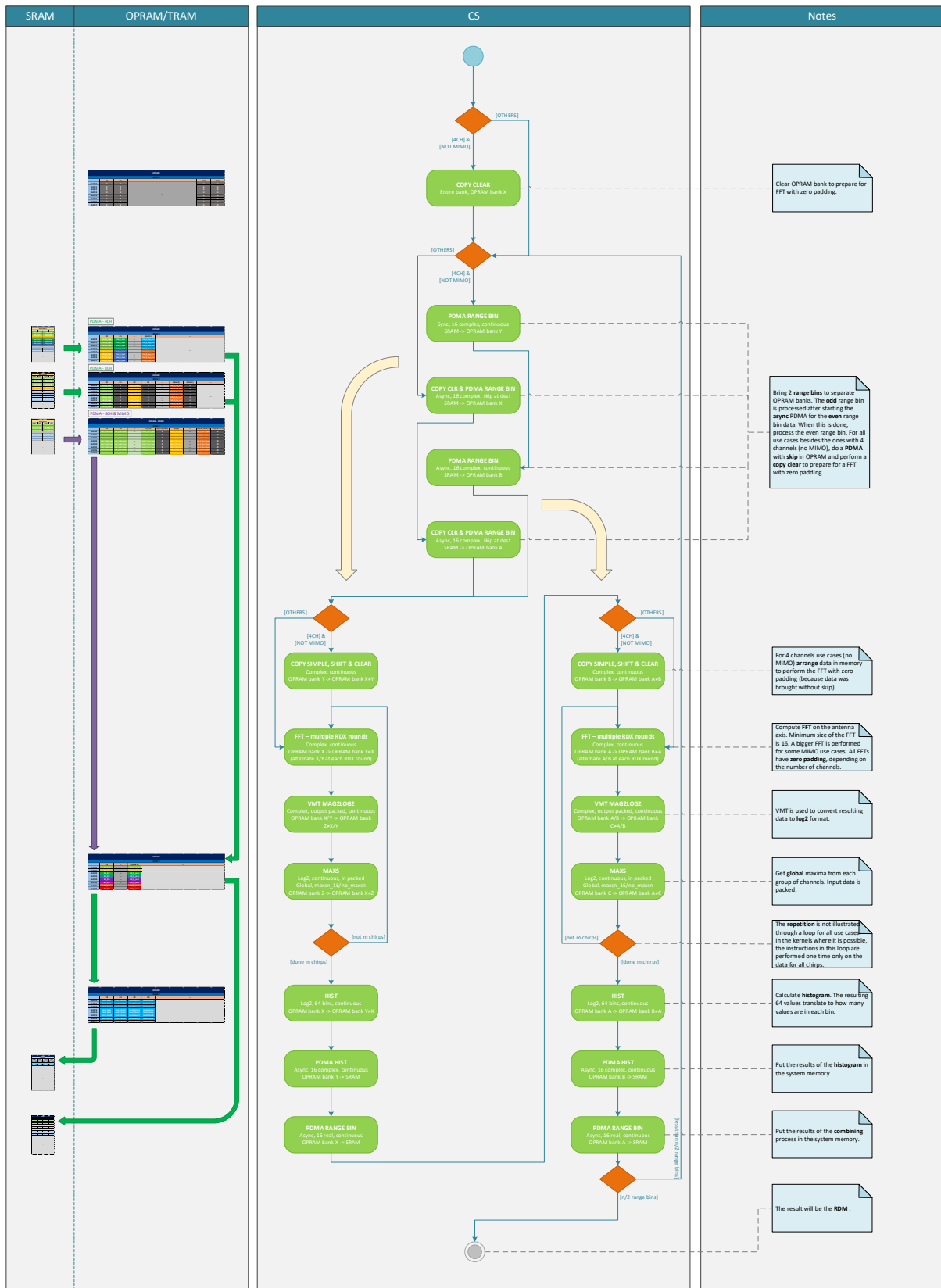
3.2.2 8CH



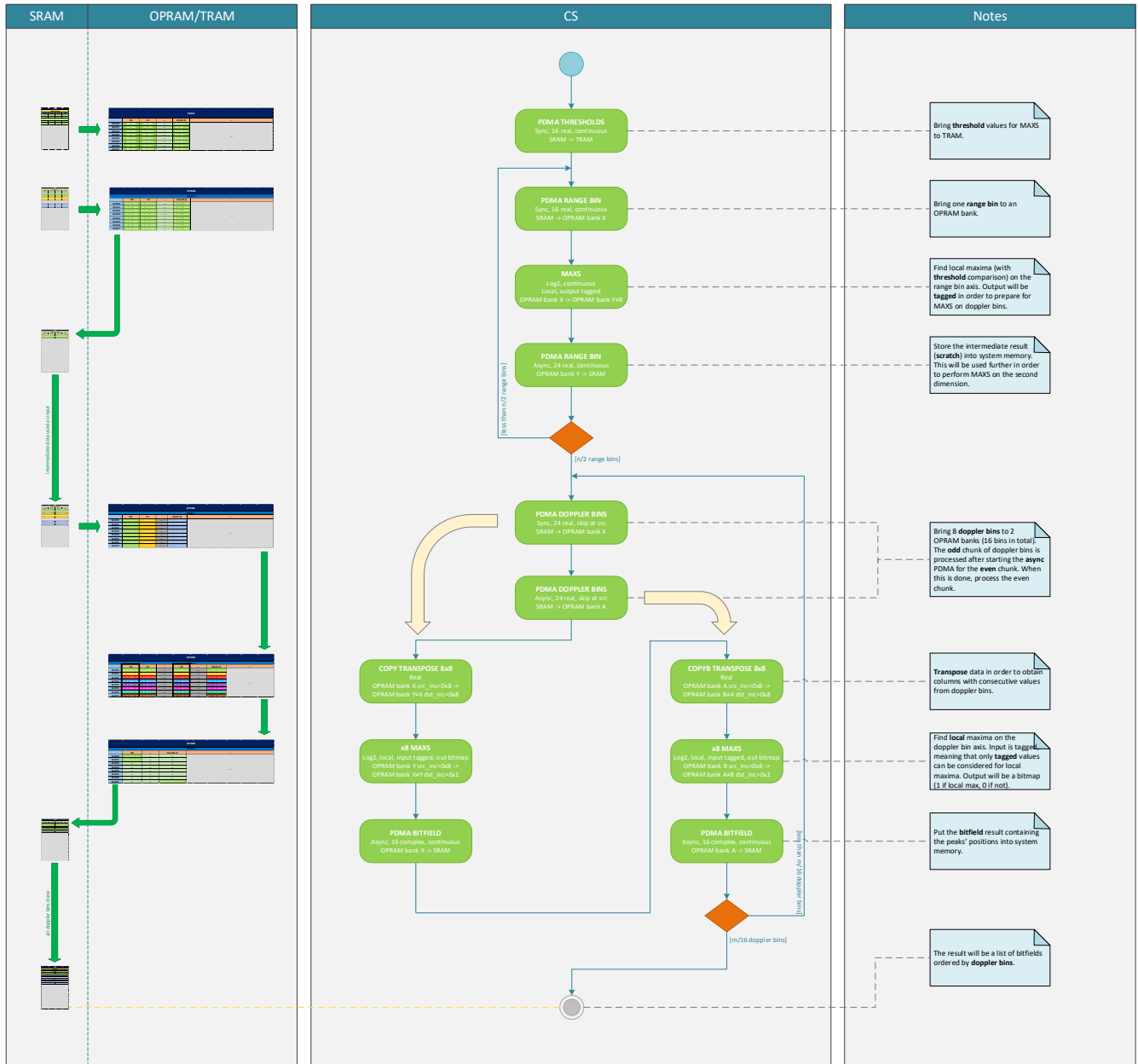
3.4 Ncc Kernel



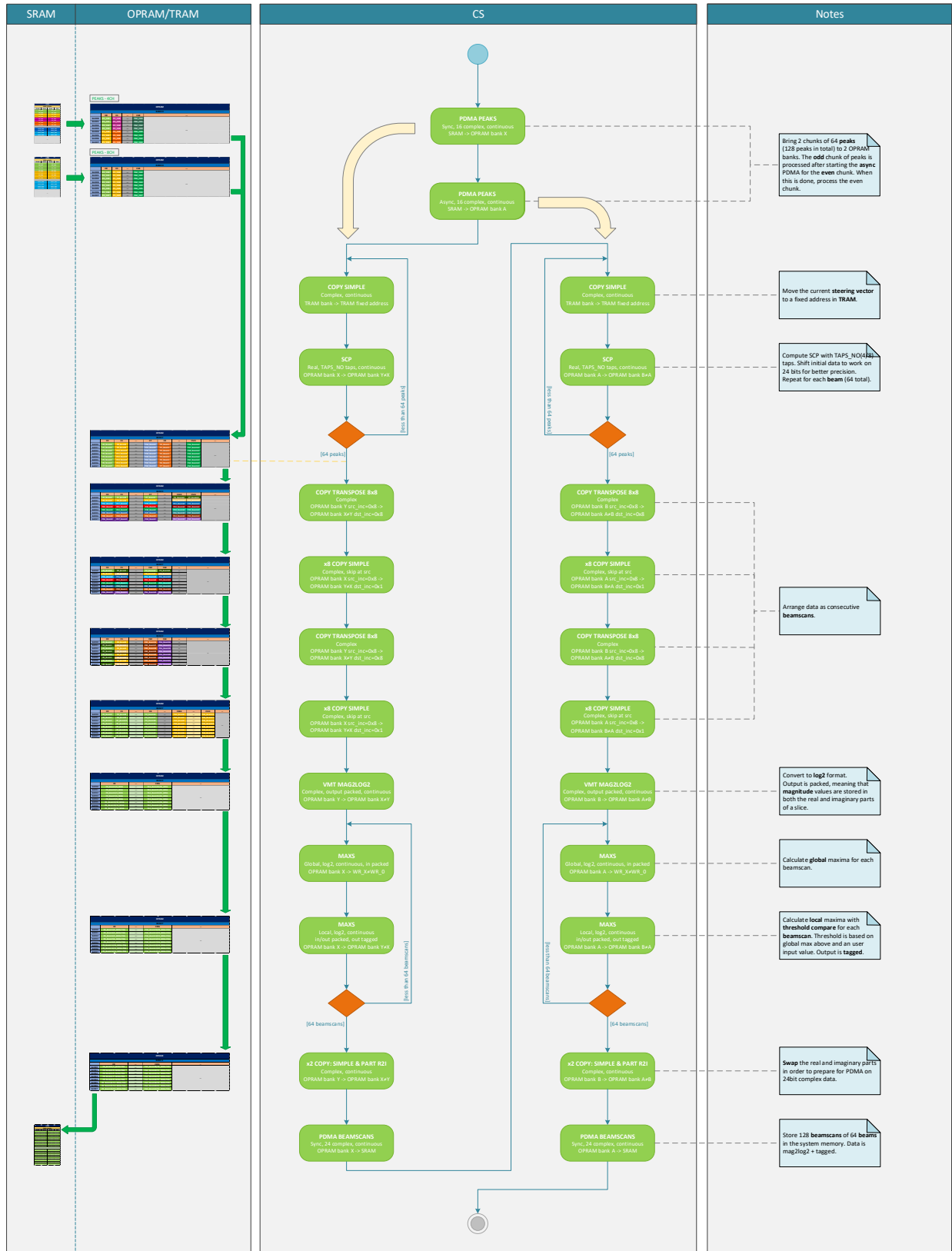
3.5 3D-FFT Kernel



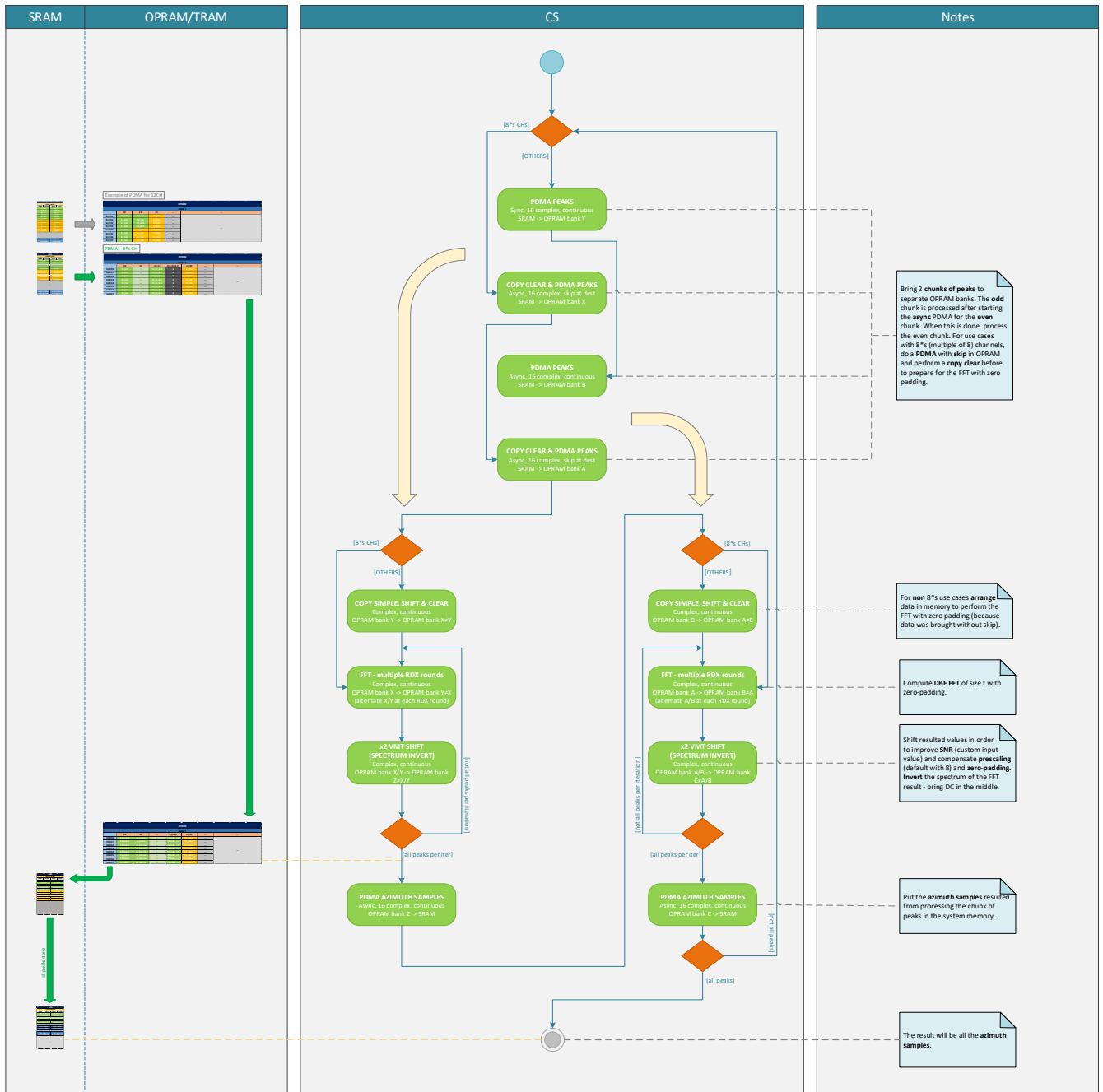
3.6 2D-PS Kernel



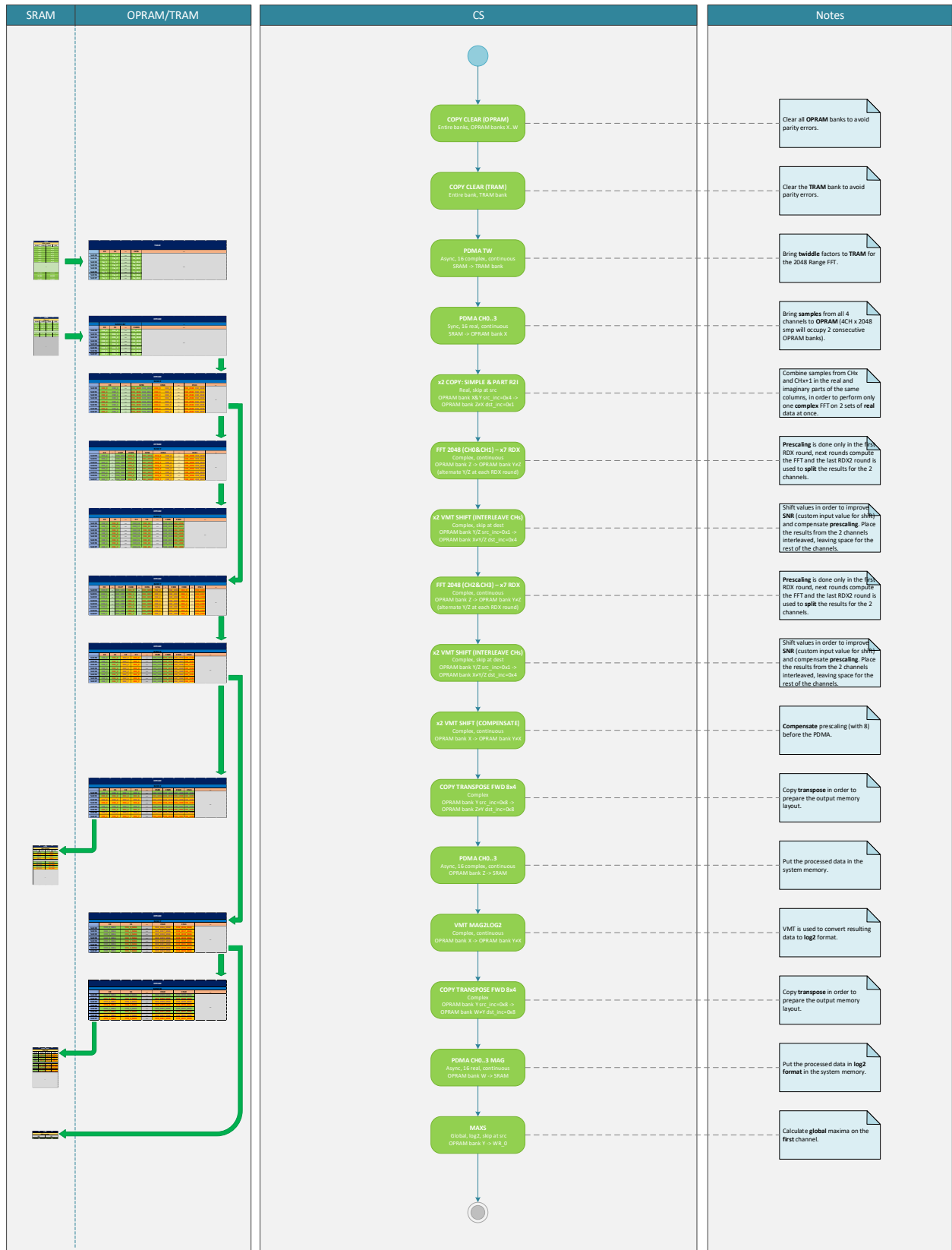
3.7 DbfDoa Kernel



3.8 DbfFFT Kernel



3.9 RfBist Kernel



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4 Document information

4.1 Abbreviations and terminology

Table 1: Abbreviations and terminology

Abbreviation	Description
RRU2	Race Runner Ultra 2
SPT	Signal Processing Toolbox
CS	Command Sequencer
OPRAM	Operand RAM
TRAM	Twiddle RAM
FFT	Fast Fourier Transform
Ncc	Noncoherent Combining
PS	Peak Search
Dbf	Digital Beamforming
Doa	Direction of Arrival
RfBist	RF Build In Self Test
MIMO	Multiple-Input Multiple-Output
CP4D	Type of data compression supported by the SPT

4.2 Revision history

Table 2: Revision history

Revision	Date	Description	Author
1.0	04-13-2021	Initial version	Adriana Ionita

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